UNWRAP: Unpacking Whether Replacement Alternatives Perform

A paired meta-analysis of packaging life cycle assessments under Indian collection rates and an honest functional unit

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HIGHLIGHTS

- 31 within-study pairs from 16 LCAs: replacements for plastic emit 1.38x as much on average and win in only 42% of pairs.
- Cartons beat plastic in all 9 pairs (0.63x); reusable glass is near parity; single-use glass and aluminium lose.
- Values taken from an industry-compiled table make alternatives look about 26% better than original sources.
- Under Indian collection rates, the aluminium can matches PET only above about 89% collection.
- Scoring packs per kg of product actually used reverses the canister vs refill-pouch tie and narrows the sachet penalty.

ABSTRACT

Background. Replacing plastic packaging with glass, aluminium, cartons or refills is widely assumed to cut carbon, yet life cycle assessments (LCAs) disagree, assume European waste systems, and divide impact by product filled rather than product used.

Methods. We screened 90 records and extracted 46 paired comparisons from 23 studies. Within-study log response ratios of global warming potential (GWP) were pooled with equal study weights (t and bootstrap intervals), with class-adjusted moderator tests, leave-one-out and funding checks. A Monte Carlo screening model recomputed beverage containers under European and Indian collection rates. A second model scored four shampoo packs per kg of product actually used, using measured residue rates.

Results. In the core pool (31 pairs, 16 studies) the alternative had 1.38 times the GWP of plastic (95% CI 1.03 to 1.88) and was better in 13 pairs. Cartons won every pair (0.63x, 0.52 to 0.73), reusable glass was near parity (0.91x), and single-use glass (2.58x) and aluminium (2.45x) lost. Values from a second-hand industry table were more favourable to alternatives (class-adjusted shift 0.74, 0.57 to 0.96), and in 4 of 5 industry-linked studies the sponsor's material ranked best. With Indian rates, the aluminium can's penalty against PET fell from 1.73x to 1.16x, with parity at 89% collection (83% to 96%). Per kg used, every shampoo pack's footprint rose by 38% to 121%; a 2 L canister that tied a refill pouch per kg filled lost to it by 1.40x (1.03 to 1.93).

Conclusions. The carbon benefit of replacement packaging is format-specific and conditional on the waste system and the functional unit. Claims should be judged per unit of product actually used, under local collection rates, and from independent sources.

KEYWORDS life cycle assessment; meta-analysis; FMCG packaging; log response ratio; India; recycling; functional unit; product residue

1. Introduction

Packaging policy increasingly favours replacing plastic with glass, aluminium, paper-based cartons or refill systems. The carbon case for these switches rests on LCAs, but published LCAs reach conflicting conclusions. A review of 53 comparative studies found that plastics were often not the worst option and that end-of-life (EoL) assumptions shaped the ranking (Dolci et al., 2025). The most comprehensive prior synthesis pooled 131 packaging LCAs but compared format averages across studies and covered only North America and Europe (Bher and Auras, 2024).

Three gaps motivate this study. First, comparisons are rarely pooled *within* studies, where both packs share one system boundary, database and set of assumptions; cross-study averages mix method differences into the comparison. Second, results are almost always computed for European or US waste systems. India differs in both directions: TERI (2022) reports collection-for-recycling rates of 85% for aluminium cans, 70% for PET, 60% for glass and 54% for cartons, while

open burning of uncollected waste remains common (Cottom et al., 2024). Third, every packaging LCA we found uses product *filled* as the functional unit. Product that stays in the pack or is never used is invisible, even though measured residues reach about 7% for pump dispensers and 10% for viscous products in tubes (Schinkel et al., 2023; Sustainability, 2025).

We test three hypotheses. **H1:** replacement packaging has lower GWP than the plastic it replaces. **H2:** any advantage shrinks or reverses under Indian collection rates. **H3:** scoring packs per unit of product used, not filled, changes their ranking.

2. Methods

2.1 Search, eligibility and screening

We searched Google Scholar, Scopus, ScienceDirect and grey literature (UNEP Life Cycle Initiative, TERI, trade and industry LCAs) using the string ("life cycle assessment" OR LCA) AND packaging AND (sachet OR refill OR glass OR PET OR "mono-material" OR multilayer OR reusable OR carton) AND (comparative OR comparison), and snowballed backward from reviews. Studies were eligible if they (i) assessed FMCG primary packaging, (ii) were published from 2005 onward or were tabulated by an eligible review, and (iii) reported 100-year GWP for a conventional and an alternative pack under one functional unit and method. Reviews, transport packaging, medical packaging and marketing claims without a method were excluded. Of 90 records, 38 were excluded at title and abstract, 52 were assessed, 8 lacked paired numbers and 20 were eligible but had no openly accessible numbers (Fig. 1). The final set holds 24 reports from 23 studies, yielding 46 pairs and 74 distinct GWP values (Appendix Table A1).

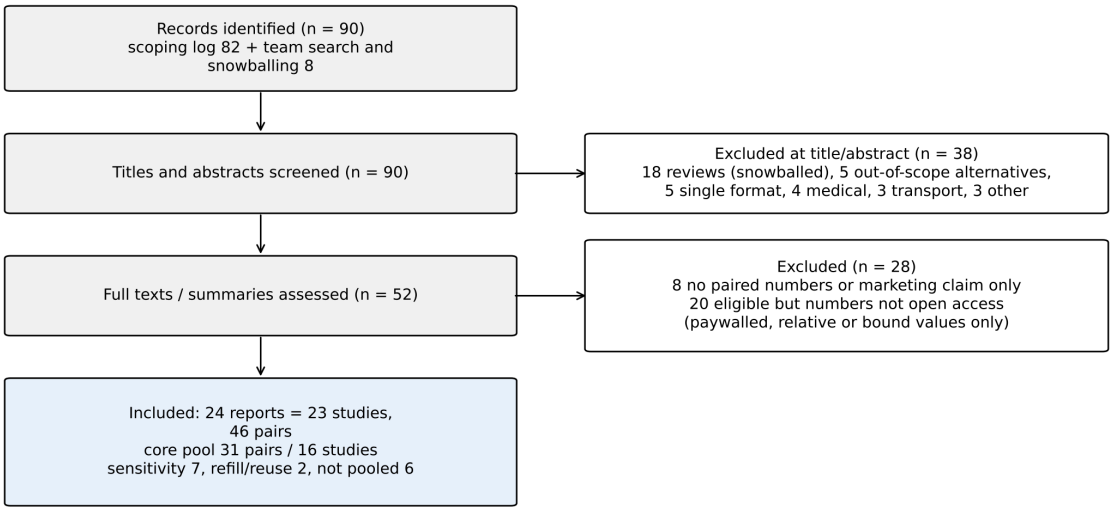


Fig. 1. Flow of records through screening. Scoping counts come from an itemised search log (workbook sheet S5).

2.2 Data extraction, coding and verification

For each pair we extracted the two GWP values, functional unit, product, region and location in the source, and coded the material class, funder (academic, industry-linked, other or unclear), system boundary, functional-unit basis, data source and whether EoL rates were stated. Eighteen values came from a compilation commissioned by the beverage carton industry (Wohner and Tacker, 2020); all were checked against its Tables 1 to 4. Primary values for Stefanini et al. (2021), Kim and Park (2020) and Kröhnert and Stucki (2021) were checked against the papers. A Japanese shampoo study whose source could not be traced was moved to sensitivity analysis.

The **core pool** comprises pairs with a plastic baseline, a true format switch and the same product (31 pairs, 16 studies). Seven pairs enter sensitivity runs only: two with a likely unit error (ratio 58x; Saleh), two that compare different beverages (Pasqualino et al., 2011), a takeaway tray outside FMCG scope, an extra 30-use scenario, and the untraced study. Two same-material refill or reuse pairs and six pairs without a plastic baseline are reported but not pooled (Appendix Fig. A1).

2.3 Stage 1: paired meta-analysis

For pair *j* in study *i* the effect size is the log response ratio

$$y_{ij} = ln(\ GWP_{alt,ij} / GWP_{conv,ij}\ ) \tag{1}$$

so that a pooled ratio exp(μ) below 1 favours the alternative. Pairs were averaged within studies to give the study mean $\bar{y}_i$, and studies were pooled with equal weight because within-study variances are not reported:

$$\hat{\mu} = (1/k)\ \Sigma_i\ \bar{y}_i, \quad \hat{\tau} = SD(\bar{y}_i), \quad CI = \hat{\mu} \pm t_{k-1}\ \hat{\tau}/\sqrt{k} \tag{2}$$

We report this t-interval, a study-level bootstrap interval (20,000 resamples; the headline interval) and a 95% prediction interval $\hat{\mu} \pm t_{k-1}\hat{\tau}\sqrt{(1 + 1/k)}$. Moderators were tested on class-centred residuals (each pair minus its material-class mean), so that method effects are not confounded with material mix; differences between moderator levels were bootstrapped within levels. Robustness was assessed with twelve sensitivity subsets and leave-one-study-out. Funding bias was assessed by subset analysis and by recording whether the sponsor's own material ranked best. Funnel plots and regression tests for small-study effects require standard errors and were therefore not possible.

2.4 Stage 2: beverage containers under collection rates

A screening model gives GWP per litre filled for a 500 mL aluminium can, a 600 mL PET bottle and a 650 mL single-use glass bottle, using TERI (2022) masses m, distances D and recycling yields y:

$$G_{Al} = m(V{+}K) - mcy(V{-}S) + mDe; \quad G_{glass} = mV - mcy\Delta V + mDe; \quad G_{PET} = m(V{+}K) - mcyq(V{-}S) + m(1{-}c)bB + mDe \tag{3}$$

where V and S are virgin and secondary material factors, K conversion overheads, c the collection rate, q the quality of the open-loop PET credit, b the burned share of uncollected PET, B = 2.29 kg CO_2/kg and e the truck emission factor. Factors come from IAI, the LSE Grantham Institute, FEVE and Hopewell et al. (2009); the three unsourced overheads and b (10% to 80%) were varied over stated ranges (workbook S6). Monte Carlo simulation (20,000 draws, triangular distributions) was run for European rates (aluminium 71%, PET 48%, glass 80%) and Indian rates (TERI; PET also at 90%). The parity rate c* solves $G_{Al}(c^*) = G_{PET}$.

2.5 Stage 3: the honest functional unit

We modelled four common shampoo packs: a 6 mL multilayer sachet, a 200 mL pump bottle, a 500 mL refill pouch and a 2 L canister. For a pack with fill F, pack GWP G, residue share r, unused share u and product footprint P (kg CO_2e per kg product), the conventional and honest metrics are

$$E_{filled} = G / F, \quad E_{used} = (G + PF) / [F(1{-}r)(1{-}u)] - P \tag{4}$$

E_{used} charges the pack for the product it strands, so both metrics are packaging-attributable and directly comparable. Residues are measured values for one shampoo across pack types: 7.1% (200 mL pump), 2.7% (500 mL pouch) and 3.2% (2 L canister) (Sustainability, 2025). Sachet residue uses the 3% to 11% packaging-caused loss measured in 9 to 20 g portion packs (Doğdu et al., 2024). The canister also carries 0% to 5% never-used product, reflecting evidence that oversized packs drive household waste (Williams et al., 2012). Pack masses, material factors and P (1 to 5) are assumptions varied over wide ranges (Appendix Table A2). Indian EoL was applied: multilayer packs are not recycled, rigid packs are collected at 40% to 80%, and a share of uncollected waste is burned. A no-burning variant was also run.

3. Results

3.1 Pooled effect: alternatives are not better on average (H1)

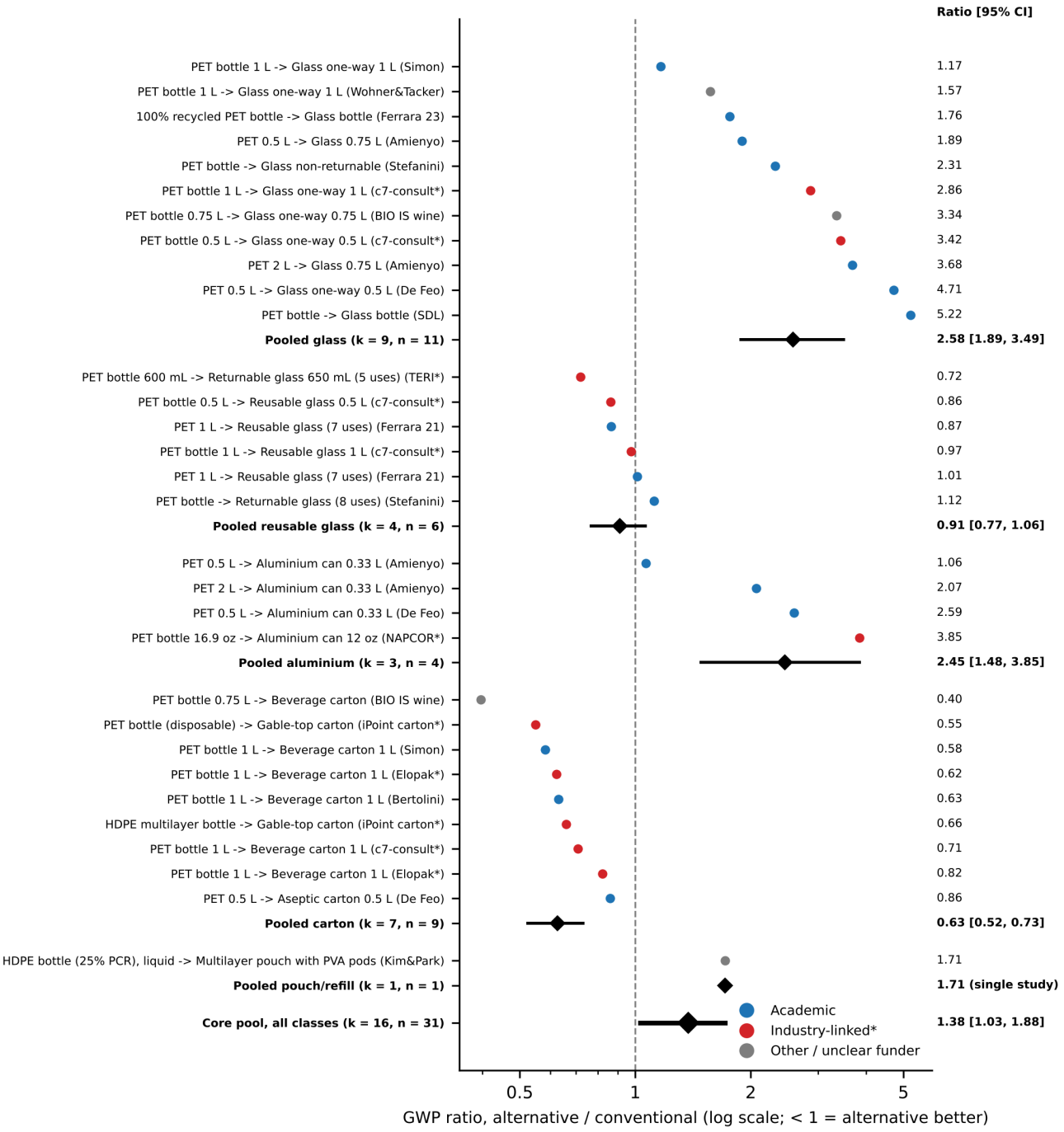


Fig. 2. Forest plot of the core pool. Points are published pairs (alternative / conventional GWP); diamonds are pooled estimates with study-level bootstrap 95% intervals. Colour indicates funder; * marks studies linked to a producer or trade body.

Table 1. Pooled GWP ratios by material class (core pool).

Alternative vs plastic	k	n	Ratio	Bootstrap 95% CI	t 95% CI	Alternative better
Single-use glass	9	11	2.58	1.89 to 3.49	1.76 to 3.79	0 of 11
Aluminium can	3	4	2.45	1.48 to 3.85	0.75 to 8.06	0 of 4
Flexible pouch (detergent pods)	1	1	1.71	single study	n/a	0 of 1
Reusable glass (5 to 8 uses)	4	6	0.91	0.77 to 1.06	0.68 to 1.22	4 of 6
Beverage carton	7	9	0.63	0.52 to 0.73	0.50 to 0.78	9 of 9
Core pool, all classes	**16**	**31**	**1.38**	**1.03 to 1.88**	**0.98 to 1.92**	**13 of 31**

k = studies, n = pairs. Core pool: $\tau = 0.63$ (ln units); 95% prediction interval for a new study 0.35 to 5.47.

Across the core pool the alternative had 1.38 times the GWP of the plastic pack (bootstrap 95% CI 1.03 to 1.88) and was better in 13 of 31 pairs, so H1 is not supported. Heterogeneity is large ($\tau = 0.63$) and is largely explained by material. Cartons beat plastic in every pair (0.63x) and reusable glass was near parity (0.91x, 0.77 to 1.06). Single-use glass never beat plastic (2.58x) and aluminium lost in all four pairs (2.45x). Excluding single-use glass alone moves the pooled ratio to 0.96 (0.71 to 1.35). The one flexible-format pair (detergent pods in a multilayer pouch) was worse than the rigid bottle, mainly because of the water-soluble PVA film. Same-material refill and reuse pairs, which were not pooled, favoured reuse: a shampoo refill cut packaging GWP by about 70% against a new glass bottle (Kröhnert and Stucki, 2021), and a returnable beer bottle by 53% after six cycles (Mata and Costa, as tabulated by Stefanini et al., 2021).

3.2 Robustness, method choices and funding

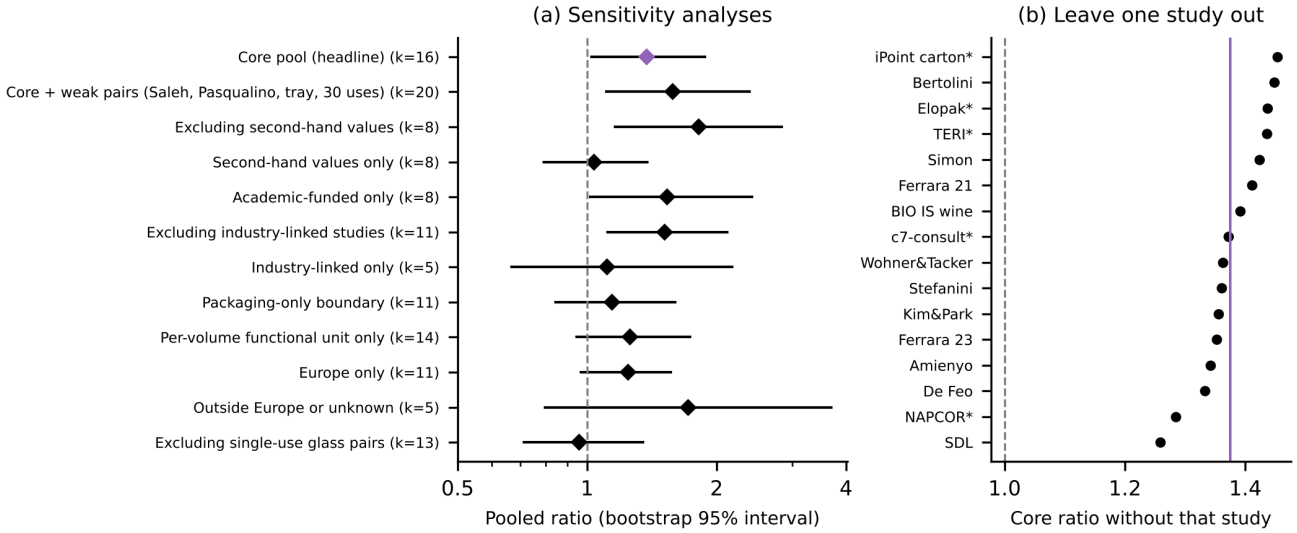


Fig. 3. (a) Pooled ratio under twelve sensitivity subsets (bootstrap 95% CI). (b) Core pooled ratio with each study removed; the vertical line is the full core estimate.

The pooled ratio stays above 1 in every subset except the one that removes single-use glass, but its interval includes 1 in 7 of the 12 subsets, so the size of the penalty is less certain than its direction. Adding the seven weak pairs raises the ratio to 1.58 (1.11 to 2.39). Leave-one-out estimates range from 1.26 (without SDL) to 1.45 (without the iPoint carton study), so no single study drives the result.

Table 2. Class-adjusted effect of coded method choices on the GWP ratio.

Level compared with all others	Studies	Raw ratio (level / others)	Class-adjusted shift	95% CI
Second-hand compiled table vs original source	8 / 8	1.04 / 1.82	0.74	0.57 to 0.96
Europe vs elsewhere or unreported	11 / 5	1.24 / 1.72	0.77	0.54 to 1.06
Per-volume unit vs per container or per use	14 / 2	1.26 / 2.57	0.77	0.55 to 1.06
Packaging-only boundary vs including product or unreported	11 / 5	1.14 / 2.06	0.87	0.59 to 1.25
EoL rates stated vs not stated	4 / 12	1.41 / 1.37	0.88	0.69 to 1.11
Industry-linked vs other funders	5 / 11	1.11 / 1.52	1.14	0.86 to 1.51

Shift = ratio for the level relative to all others after centring each pair on its material-class mean (1 = no effect; below 1 = the level makes alternatives look better). Intervals from study-level bootstrap.

The only method choice with a clear effect is data source. Values taken from the industry-commissioned compilation pooled at 1.04 against 1.82 for original sources, and remained about 26% more favourable to alternatives after removing material mix (shift 0.74, 0.57 to 0.96). European studies, per-volume units and packaging-only boundaries also lean towards the alternative (shifts 0.77 to 0.87), consistent with higher Western collection rates and narrower boundaries flattering replacements, but none is clear with 16 studies. At study level, the sponsor's own material ranked best in **4 of 5 industry-linked studies** (PET trade body: PET; two carton makers: carton; can maker: aluminium); the exception was a PET bottle maker's study in which cartons won. Academic studies alone pooled at 1.54 (1.02 to 2.42).

3.3 Indian collection rates (H2)

Table 3. Stage 2 model results per litre filled (Monte Carlo, 20,000 draws).

Collection-rate scenario	Al can / PET, median (90% interval)	Draws with Al < PET	Glass / PET
Europe (Al 71%, PET 48%, glass 80%)	1.73 (1.45 to 2.09)	0%	5.3 (4.1 to 6.9)
India, TERI rates (Al 85%, PET 70%, glass 60%)	1.16 (0.94 to 1.43)	13%	5.7 (4.6 to 7.3)
India, PET collected at 90%	1.30 (1.05 to 1.62)	2%	6.4 (5.1 to 8.3)
India, TERI rates, aluminium-favourable overheads	0.88 (0.75 to 1.05)	89%	n/a

"Draws" is the share of parameter draws, not a probability.

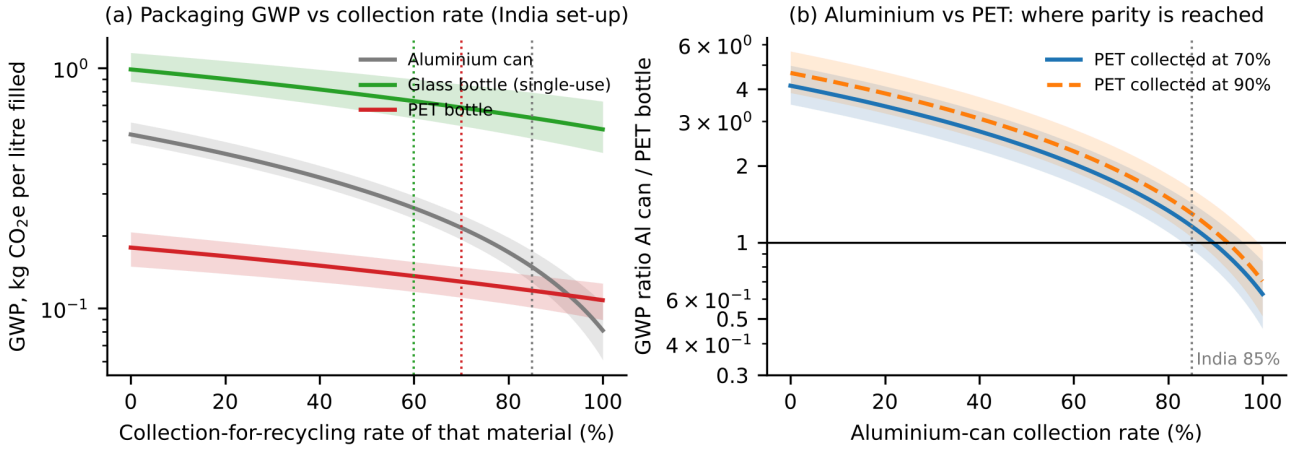


Fig. 4. (a) GWP per litre against the collection rate of each material (India set-up; bands = 5th to 95th percentile; dotted lines = Indian rates). (b) Aluminium can / PET ratio against aluminium collection; parity is reached near 89%.

Under European rates the model places aluminium at 1.73x PET and glass at 5.3x, the same direction as the Italian study by De Feo et al. (2.6x and 4.7x). Under TERI's Indian rates the aluminium penalty shrinks to 1.16x and aluminium beats PET in 13% of draws; at 90% PET collection this falls to 2%. Parity requires **c* = 89% aluminium collection** (90% interval 83% to 96%) against a reported 85%; even with the most favourable overheads c* is 81% (76% to 86%). Single-use glass remains about 5x PET at any collection rate. The aluminium to PET ratio was most sensitive to the can-making overhead (Spearman $\rho = 0.48$), followed by PET conversion, aluminium recycling yield and PET resin factor ($|\rho|$ 0.37 to 0.39). H2 is therefore not supported in its original form: India narrows the gap for aluminium but does not reverse the ranking, and glass is unaffected.

3.4 The honest functional unit (H3)

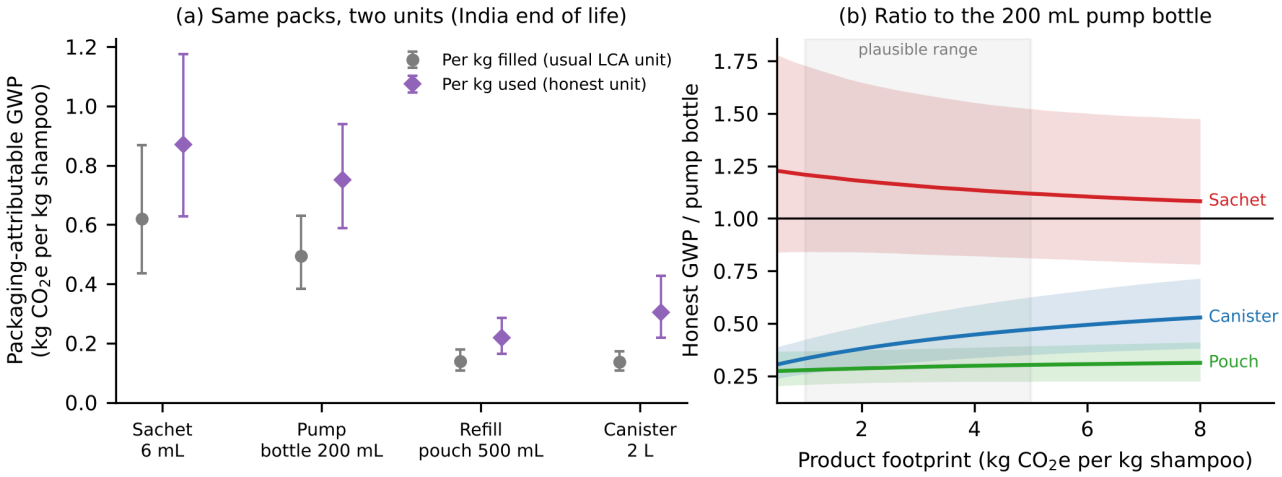


Fig. 5. (a) Packaging-attributable GWP of four shampoo packs per kg filled (grey) and per kg used (purple), Indian EoL; bars = 5th to 95th percentile. (b) Honest-unit ratio to the 200 mL pump bottle as the product footprint varies (shaded = plausible range).

Table 4. Stage 3 results, kg CO_2e per kg shampoo, median (90% interval), Indian EoL with open burning.

Pack	Residue (+ unused)	E_{filled} (usual unit)	E_{used} (honest unit)	Increase
Sachet 6 mL	3% to 11%	0.62 (0.44 to 0.87)	0.87 (0.63 to 1.18)	+38%
Pump bottle 200 mL	5.2% to 9.0%	0.50 (0.38 to 0.63)	0.75 (0.59 to 0.94)	+50%
Refill pouch 500 mL	1.1% to 4.0%	0.14 (0.11 to 0.18)	0.22 (0.16 to 0.29)	+54%
Canister 2 L	2.0% to 4.5% (+ 0% to 5%)	0.14 (0.11 to 0.17)	0.30 (0.22 to 0.43)	+121%

Without open burning all values fall by 10% to 25% and rankings are unchanged.

Counting stranded product raised every pack's footprint by 38% to 121%, often more than the difference between packs, so H3 is supported. Three results follow. **First, bigger is not always better.** Per kg filled, the 2 L canister tied with the refill pouch (ratio 0.98, 0.72 to 1.33; canister lower in 53% of draws). Per kg used it lost (1.40, 1.03 to 1.93; lower in only 4% of draws), and the pouch became the lowest-carbon pack in 96% of draws. **Second, the sachet penalty narrows but survives.** Against the pump bottle, which strands 7% of its contents, the sachet ratio fell from 1.25 (0.84 to 1.85) to 1.16 (0.83 to 1.61), and the share of draws in which the sachet wins rose from 18% to 23%. The sachet result is most sensitive to sachet mass ($\rho = 0.53$), fill and residue ($|\rho| = 0.36$). **Third, refill pouches win under both units**, at about 0.3x the pump bottle. Beverage residues are below 1% (Schinkel et al., 2023), so the honest unit leaves the Stage 1 beverage results unchanged.

4. Discussion

Principal findings. Replacing plastic does not reduce carbon in general. The benefit depends on the format: cartons reliably beat plastic, reusable glass matches it only after several uses, and single-use glass and aluminium usually lose. This agrees with Dolci et al. (2025), who report that single-use PET beats refillable glass below roughly 10 reuses, consistent with the 5 to 8 use pairs here sitting at parity. The pairing design removes much of the method noise that makes cross-study comparisons hard to read, and the remaining heterogeneity is largely explained by material class.

Context dependence. Indian collection rates help aluminium more than any other material, but parity still requires near-complete collection, so the advantage is fragile rather than established. For aluminium, investment in collection moves the result more than lightweighting; for glass, reuse rather than recycling is the lever (TERI, 2022, found that five refills cut glass GWP by 64% to 73%).

Evidence quality. Two signals call for caution when published LCAs inform policy. Sponsored studies tended to favour the sponsor's material, and values passed through an industry-compiled table were more favourable to alternatives than original sources. Neither proves bias in any single study, but both argue for independent, open comparisons with stated EoL assumptions.

The functional unit. The honest-unit test shows that the conventional functional unit can decide which pack "wins". Product stranded in a pack can add more carbon than the pack itself, which favours packs that empty well (pouches with spouts, right-sized packs) over simply larger ones. It also implies that restricting sachets gives a carbon gain only if users move to refills rather than to small pump bottles. We found no packaging LCA or meta-analysis that reports this adjustment.

Limitations. (i) Only 24 of 44 eligible reports had openly accessible numbers, short of the 25 to 40 studies and 60 pairs targeted; the excluded reports (notably the IFEU beverage LCAs) could change class estimates. (ii) Eighteen core values come from one industry-commissioned compilation; they match that compilation, but not every original paper could be opened. (iii) Within-study variances are not reported, so studies are equally weighted, intervals are descriptive, and small-study bias could not be tested formally. (iv) No paired sachet LCA with usable numbers exists and no mono-material comparison met the criteria, so Stage 3 is a model built on measured residues and assumed pack masses, and its sachet residue is a proxy from food portion packs. (v) The Stage 2 model omits secondary packaging, uses three unsourced overheads and overstates glass relative to aluminium in India (4.9x against TERI's 3.5x), so c* is indicative. (vi) Only GWP was assessed; other impacts such as marine litter can rank packs differently (Stefanini et al., 2021).

Future work. Extract the IFEU and other paywalled LCAs; obtain within-study uncertainty to enable inverse-variance weighting and funnel tests; run measured residue trials on Indian sachets and bottles; and extend the honest unit to food and home-care products.

5. Conclusions

Across 31 published within-study comparisons, replacements for plastic were lower-carbon in only about two of five cases. Cartons won consistently; single-use glass and aluminium did not. Under Indian collection rates the aluminium can approaches PET but does not match it unless about 89% of cans are collected. Measuring per kg of product actually used, rather than per kg filled, changes which large pack is best and narrows the sachet penalty. Packaging claims and policy should be judged per unit of product used, under the local waste system, and on independent evidence.

Data and code availability

All extracted data, coding decisions and results are in *UNWRAP_data_and_results.xlsx* (sheets S1 to S15, with live formulas for ratios, pooled t-intervals and base-case models). The full analysis is reproducible with *analysis.py* and *build_workbook.py* from *paired_comparisons.csv* and *prisma_log.csv* (Python 3 with numpy, pandas, scipy, matplotlib and openpyxl; fixed random seeds).

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Further sources (Saleh; Mata and Costa; SDL thesis; iPoint; LSE Grantham Institute; FEVE; CSIR-NCL/PACE; Japanese shampoo LCA) are listed with links in workbook sheet S15.

Appendix A. Supplementary tables and figures

Table A1. Characteristics of included studies.

ID	Study	Year	Region	Product(s)	Pairs (core)	Funder	Data source	Boundary
S1	De Feo	n/r	Italy	Bottled water	3 (3)	Academic	abstract or summary	n/r
S2	Ferrara 23	2023	Italy	Olive oil	1 (1)	Academic	abstract or summary	n/r
S3	NAPCOR*	2023	USA	Water/CSD	1 (1)	Industry-linked	abstract or summary	packaging only
S4	SDL	n/r	n/r	Carbonated drinks	1 (1)	Academic	abstract or summary	n/r
S5	iPoint carton*	2021	n/r	Fresh milk	2 (2)	Industry-linked	abstract or summary	packaging only
S6	Gallego-S.	2019	Europe	Takeaway food container	1 (0)	Academic	abstract or summary	packaging only
S7	TERI*	2022	India	Beer; Beverages; Carbonated drinks	3 (1)	Industry-linked	full text	packaging only
S8	Stefanini	2021	Italy	Milk	4 (2)	Academic	full text	packaging only
S9	Pasqualino	2011	Spain	Beverages	2 (0)	Academic	compiled table	incl. product
S10	Amienyo	2013	UK	Carbonated drinks	4 (4)	Academic	compiled table	incl. product
S11	Saleh	2015	Palestine	Beverages	2 (0)	Academic	compiled table	n/r
S12	Mata&Costa	2001	Portugal	Beer	2 (0)	Academic	compiled table	n/r
S13	Ferrara 21	2021	Italy	Natural water; Sparkling water	2 (2)	Academic	full text	n/r
S14	c7-consult*	2019	Austria	Juice; Milk	5 (5)	Industry-linked	compiled table	packaging only
S15	Wohner&Tacker	2019	Austria	Milk	1 (1)	Other/unclear	compiled table	packaging only
S16	Wohner PeerJ	2019	Austria	Milk	1 (0)	Other/unclear	compiled table	packaging only
S17	Simon	2016	Hungary	Beverages	2 (2)	Academic	compiled table	packaging only
S18	Bertolini	2016	Italy	ESL milk	1 (1)	Academic	compiled table	packaging only
S19	BIO IS wine	2010	Nordic countries	Wine	2 (2)	Other/unclear	compiled table	packaging only
S20	Elopak*	2016	Europe	Juice (aseptic); Juice (chilled)	2 (2)	Industry-linked	compiled table	packaging only
S21	Kim&Park	2020	North America	Laundry detergent	2 (1)	Other/unclear	full text	packaging only
S22	Krohnert&Stucki	2021	Switzerland	Shampoo	1 (0)	Academic	full text	packaging only
S23	Japan shampoo	2021	Japan	Shampoo	1 (0)	Other/unclear	abstract or summary	packaging only

* linked to a producer or trade body of a compared material. n/r = not reported. Data source describes how values were obtained for this review.

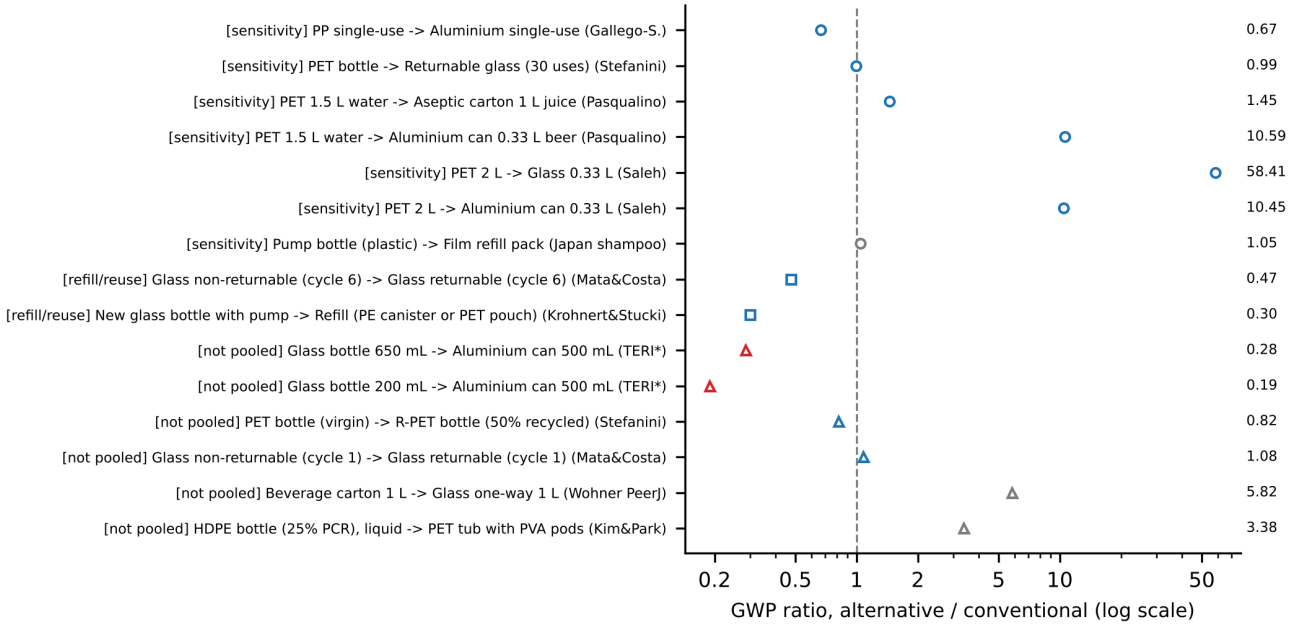


Fig. A1. Pairs outside the core pool: sensitivity-only pairs (circles), same-material refill or reuse pairs (squares) and pairs without a plastic baseline or not a format switch (triangles).

Table A2. Stage 3 model inputs (triangular distributions: low, mode, high).

Symbol	Low	Mode	High	Unit	Status	Basis
P_prod	1	2.5	5	kg CO2e/kg shampoo	assumed	Cradle-to-gate shampoo formulation; no single source, wide range tested
e_lam	3	4	5.5	kg CO2e/kg	assumed	Multilayer laminate (PET/Al/PE) incl. conversion and print
e_rigid	2.4	2.8	3.5	kg CO2e/kg	assumed	HDPE/PP bottle incl. moulding; virgin polyolefin about 1.9 plus conversion
e_can	2.2	2.5	3	kg CO2e/kg	assumed	HDPE canister incl. moulding (lighter conversion per kg)
dV_rigid	1	1.3	1.6	kg CO2e/kg	assumed	Virgin minus recycled polyolefin
q_rigid	0.3	0.5	1	fraction	assumed	Credit quality, open loop (as q_PET in Stage 2)
y_rigid	0.85	0.85	0.85	fraction	assumed	Recycling yield (as PET in TERI 2022)
c_rigid	0.4	0.6	0.8	fraction	assumed	Collection of rigid HDPE/PP in India
burn_lam	2.3	2.6	3.1	kg CO2/kg	derived	Fossil carbon of PET (2.29) to PE (3.14) layers
burn_po	3.14	3.14	3.14	kg CO2/kg	derived	Polyethylene 85.7% C x 44/12
b_IN	0.1	0.5	0.8	fraction	assumed	Share of uncollected waste open-burned (as Stage 2)
m_sachet	0.5	0.7	1	g per pack	assumed	6 mL multilayer sachet
V_sachet	5	6	8	g per pack	assumed	Typical Indian shampoo sachet 5 to 8 mL
m_bottle	25	32	40	g per pack	assumed	200 mL bottle plus pump
m_pouch	10	13	16	g per pack	assumed	500 mL stand-up refill pouch
m_can	80	100	120	g per pack	assumed	2 L HDPE canister
r_sachet	0.03	0.06	0.11	fraction	sourced (proxy)	R05: 9 to 20 g portion packs, 3 to 11% product left due to packaging
r_bottle	0.052	0.071	0.09	fraction	sourced	R01: shampoo in 200 mL pump 7.14%; pumps mean 5.23%
r_pouch	0.011	0.027	0.04	fraction	sourced	R01: shampoo 500 mL pouch 2.67%; with spout 1.08%
r_can	0.02	0.032	0.045	fraction	sourced	R01: shampoo 2000 mL canister 3.17%
u_can	0	0.02	0.05	fraction	assumed	Product never used from very large packs (pack-size waste, R06 to R08)